

Aerospace series

**Standard bulb angles, extruded
Aluminium alloy 7175
with peripheral coarse grain control
 $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$
Dimensions**

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1 Scope

This standard specifies the dimensions, tolerances and mass of extruded bulb angles in aluminium alloy 7175 with peripheral coarse grain control, $1,2\text{ mm} \leq a \leq 38,0\text{ mm}$, for aerospace applications.

2 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS03-05-000	Aluminium and aluminium alloy wrought products bar and section.
AIMS03-05-009	Aluminium alloy (7175), solution treated, controlled stretched and artificially aged (T73511)extruded section $1,2\text{ mm} \leq a \leq 38,0\text{ mm}$ with peripheral coarse grain control.
AIMS03-05-012	Aluminium alloy (7175), solution treated, controlled stretched and artificially aged (T79511) extruded section $1,2\text{ mm} \leq a \leq 38,0\text{ mm}$ with peripheral coarse grain control.
AIMS03-05-019	Aluminium alloy (7175), solution treated and artificially aged (T73) extruded section $1,2\text{ mm} \leq a \leq 38,0\text{ mm}$ with peripheral coarse grain control.
DIN1748 Part 4	Extruded profiles of wrought aluminium alloys, tolerances.
EN2066	Extruded section in aluminium and aluminium alloys - General tolerances. ¹

3 Form

See figures 1, 2 and 3.

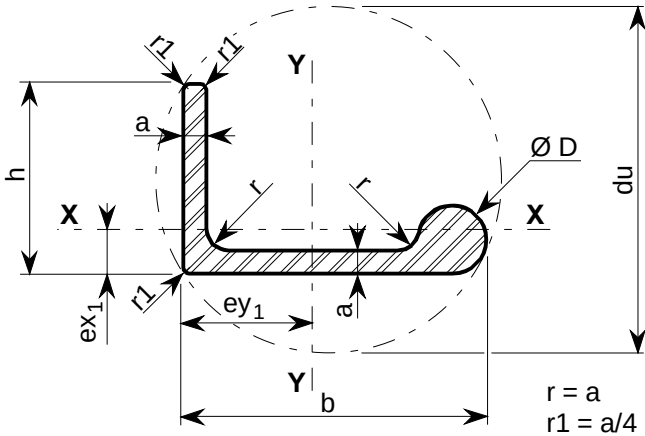


Figure 1: Standard configuration

¹ Published as ASD Standard at the date of publication of this standard

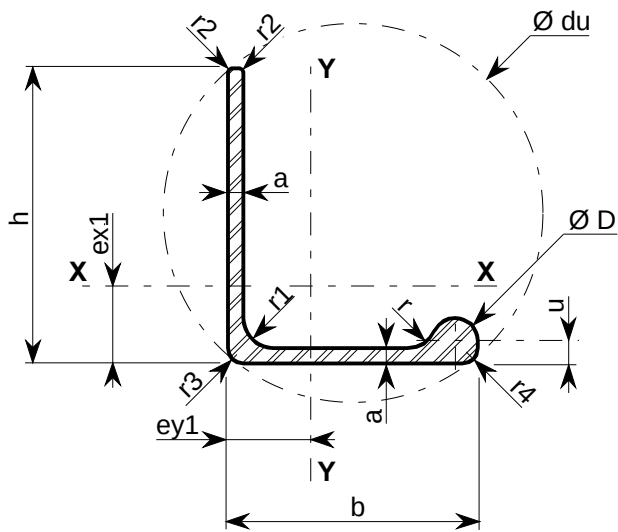


Figure 2: Configuration code "LA"

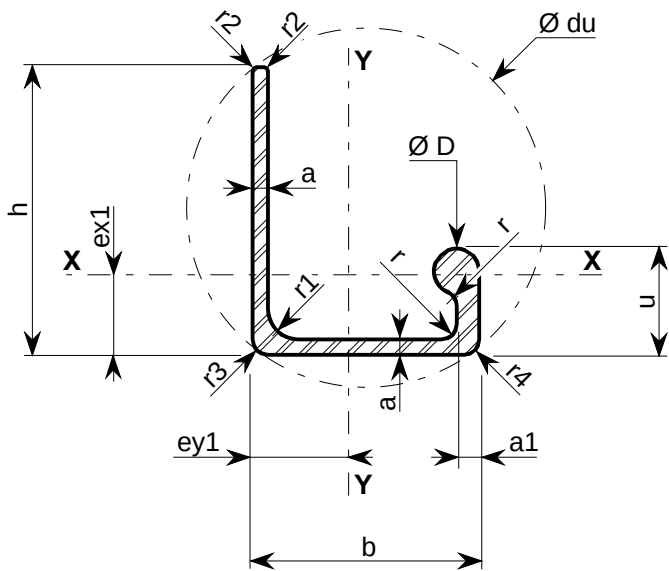


Figure 3: Configuration code "LB"

4 Recommended dimensions, mass and static values

4.1 Dimensions, cross-sections and mass

See table 1.

Table 1: Dimensions, cross-sections and mass

Dimensions in mm.

Size code	Code Form	h	b	a	a1	r	r1	r2	r3	r4	u	Ø D	Cross-section (mm ²)	Length	Linear mass ¹⁾ (kg/m)
0044	No code	16	16	1,2	-							3,5	44,00	Stock length not less than 2 000 and exact length by agreement	0,12
0062				1,6								5,0	62,00		0,17
0048			20	1,2								3,5	48,00		0,14
0069				1,6								5,0	69,00		0,19
0077			25										77,00		0,21
0084		23	18	2,0								4,0	84,00		0,23
0106		20	25									6,0	106,00		0,30
0132		24	16	1,6									82,00		0,23
0136		25	25	2,5									136,00		0,38
0120		20	32	2,0									120,00		0,34
0154		25		2,5									154,00		0,41
0167												7,5	167,00		0,47
0187		40	3,0	9,0									187,00		0,52
0234												234,00	0,65		
0242				3,2								242,00	0,68		
0285		32	50	3,0								9,0	285,00		0,80
0297				3,2									297,00		0,83
0391				4,0								12,0	391,00		1,10
0475				4,0									475,00		1,30
0614		40	63	5,0								15,0	614,00		1,70
0133	LA	29	29,4	2,0		6,0	4,0	0,5	2,0	3,0	3,0	6,0	133,40		0,38
0153		39	28			4,0				2,0			153,65		0,43
0164			33										163,65		0,46
0183	LB	38	30				3,0				2,0		14,0		183,20

¹⁾ For information, calculated with a density of 2,8 kg/dm³

4.2 Static values

See table 2.

Table 2: Static values

Dimensions in mm.

Size code	Form code	Ø du	$\frac{\text{Ø du}}{a_{\min.}}$	ex1	ey1	Ix mm ⁴	Iy mm ⁴
0044	No code	21,91	18,25	4,10	5,80	930	1 360
0062		21,61	13,50	4,30	6,40	1 270	1 940
0048		24,91	20,75	3,80	7,60	920	2 270
0069		24,61	15,38	4,00	8,30	1 260	3 450
0077		26,77	16,73	3,60	10,60	1 330	6 110
0084		28,20	14,10	6,90	5,40	4 189	2 877
0106		30,83	15,41	5,00	10,20	3 070	8 230
0136		34,13	13,65	6,70	9,40	4 896	14 062
0120		36,66	18,33	4,50	13,40	3 161	15 440
0132		27,53	17,20	6,72	6,03	3 904	2 616
0133	LA	39,95	19,97	7,21	10,75	9 475	15 121
0153		47,15	23,57	10,78	9,00	21 894	15 019
0154	No code	39,40	15,76	6,20	11,80	7 638	19 139
0164	LA	50,08	25,04	10,18	10,95	22 796	22 823
0167	No code	39,14	15,65	6,10	13,00	7 540	21 290
0183	LB	47,45	23,72	10,51	12,69	20 362	25 764
0187	No code	45,82	18,32	5,60	16,80	8 000	37 610
0234		45,56	15,18	5,90	17,00	9 322	46 021
0242		45,53	14,23	6,00	17,20	9 820	46 930
0285		52,65	17,55	7,10	20,30	20 186	80 536
0297		52,67	16,46	7,20	20,40	21 360	91 510
0391		52,38	13,08	7,80	21,80	25 680	121 380
0475		72,45	18,11	9,10	26,30	52 200	237 110
0614		71,94	14,38	9,70	27,50	62 860	299 910

5 Tolerances

5.1 Dimensional tolerances

See table 3.

5.2 Geometric tolerances

See table 3.

Table 3: Dimensional and geometric tolerances

Standard No.	Tolerance class	Code letter	
EN2066	Standard	E	Use only for order designation
DIN1748 Part 4		N	
	Precision	F	If this class is required, the code letter F has to be used for standard designation and order designation

6 Material

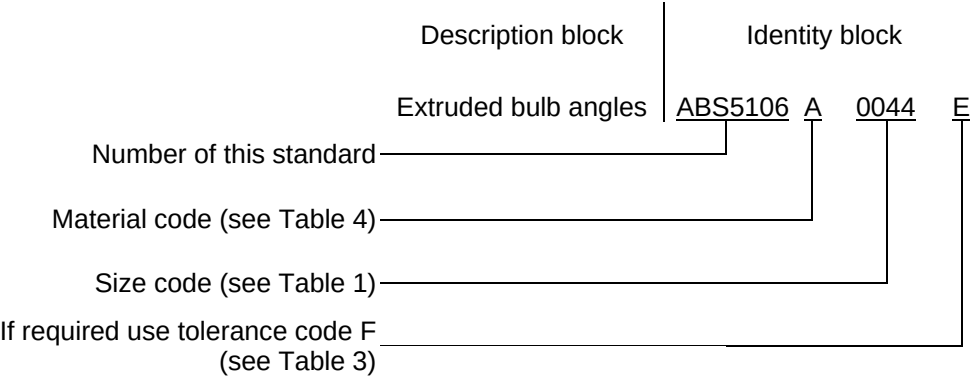
See table 4.

Table 4: Material and condition of treatment

Thickness range	Material					
	delivered	use				
	AIMS spec. No.	Condition of treatment	Code	AIMS spec. No.	Condition of treatment	Code
1,2 mm ≤ a ≤ 38,0 mm	03-05-009	T73511	A	03-05-009	T73511	A
				03-05-019	T73	B
	03-05-019	O	F			
		F	G			
	03-05-012	T79511	C	03-05-012	T79511	C

7 Designation

This type of standard shall be designated according to the philosophy of the following example:



8 Technical specification

AIMS03-05-000.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 05/96		New Standard.
2 09/01		Standard fully amended. Size code 0154 added.
3 06/03		Size code 0084 added.
4 07/10		The size codes 0153, 0164, 0133 and 0183 have been added (A350 request). The figures 2 and 3 have been added. The size code 0132 has been added (A400M request). The AIMS03-05-020 has been cancelled in the table 4.

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